

Operating Systems 2023 Spring Midterm Exam1 (2023/03/22)

1. (6%) A paging system uses a 2-level page table with an 85% TLB hit ratio. Assume that the TLB search time is ignorable comparing with the memory access time. What is the effective average memory access time comparing with the memory access time?

Ans. The effective average memory access time is the 1.3 times of the memory access time.

TLB hit = no page table lookup is needed, # main memory accesses = 1

TLB miss = need to look up address via page table, we have a 2-level table

(1) look up in top-level table to find base of 2nd level table

(2) look up in 2nd-level table to find the PTE, giving us the physical frame #

(3) look up main memory $\Rightarrow$ 3 main memory accesses for a miss

Average memory access time = $(0.85 \times 1) + (0.15 \times 3) = 0.85 + 0.45 = 1.3$

2. (10%) Please describe five performance criteria to compare CPU-scheduling algorithms and express the aim of performance criteria is to maximize or minimize respectively.

Ans.

Maximize: CPU Utilization 、 Throughput

Minimize: Turnaround Time 、 Waiting Time 、 Response Time

3. (6%) Consider a disk queue holding requests to the following cylinders in the listed order: 116, 22, 3, 11, 75, 185, 100, 87.

Assume that the disk head is at cylinder 88 and moving upward through the cylinders. What is the order that the requests are serviced by using the SCAN and C-SAN scheduling algorithm respectively?

Ans.

SCAN: 100, 116, 185, 87, 75, 22, 11, 3

C-SCAN: 100, 116, 185, 3, 11, 22, 75, 87

4. (8%) Please briefly describe the Earliest Deadline First (EDF) and the Least Slack Time First (LSTF) scheduling algorithms? When there is not enough time for everything, what is the possible phenomenon for the two scheduling?

Ans. EDF: The process with earliest deadline would be run first, and preemption happens when a process with earlier deadline arrives.

LSTF: The process with the smallest slack (deadline – remaining computation) would be run first, and preemption happens when a process with smaller slack time arrives.

If there is no enough time, EDF may finish some tasks, while LSF may miss all the deadline by approximate amount.

5. (8%) Please describe when priority inversion will occurs? How to solve it?

Ans.

(a) it can occur in a multi-tasking environment when a higher-priority task is waiting for a resource (such as a lock or a semaphore) that is currently held by a lower-priority task.

(b) Priority Inheritance may be used to avoid priority inversion. It increases the priority of a thread that is holding a lock to the maximum priority of all waiting threads for the associated lock. When the lock is released, the priority goes back to its normal level.

6. (5%) Given a UNIX i-node with ten direct blocks, one single block, one double block and one triple block. Assume that the size of a block and the block address are 8Kbytes and 4 bytes, respectively. What would be the size of the largest file allowed in byte?

Ans.

Each index block can record $8\text{kbytes}/4\text{bytes}=2\text{k}$ block

Largest file size = $(10 + 2K + (2K)^2 + (2K)^3) \text{block} \times 8\text{k bytes/block} \cong 64\text{TB}$

7. (5%) How many “hello” does the following code print ?

```
main (int argc, char **argv) {  
    int i;  
    for (i=0; i < 3; i++) {  
        fork();  
        printf("hello\n", getpid());  
    }  
}
```

Ans. 14

i = 0, Process forks child, Parent & child print “hello”

i = 1, Original parent and child, each forks a process. Now there are 4 processes, each of them prints “hello”

i = 2, Each of the 4 processes forks a child process. Now there are 8 processes, each of them prints “hello”

Total “hello” messages = 2 + 4 + 8 = 14

8. (5%) Please briefly describe the Direct Memory Access (DMA)?

Ans. Direct Memory Access (DMA) is a technique used by computers to transfer data between devices without involving the CPU. DMA allows devices like hard drives, sound cards, and network cards to read or write data directly to or from the main memory, without the need for the CPU to intervene in every transfer, which would waste a significant amount of CPU time. DMA controllers can access the memory independently of the CPU and transfer large amounts of data at high speed, thereby reducing CPU usage and improving system performance.

9. (12%) Please give at least 4 differences between user-level thread and kernel-level thread.

Ans.

User-level threads and kernel-level threads are two different approaches to implement threading in an operating system. The key differences between them are as follows:

1. Management: User-level threads are managed entirely by the application, whereas kernel-level threads are managed by the operating system.
2. Scheduling: User-level threads are scheduled by the thread library or the application, while kernel-level threads are scheduled by the operating system's scheduler.
3. Context Switching: Switching between user-level threads is faster because it doesn't require the intervention of the operating system. In contrast, kernel-level threads require a context switch to move from user mode to kernel mode, which can take more time.
4. Scalability: Kernel-level threads can take advantage of multi-core processors and can run concurrently on different cores. In contrast, user-level threads are bound to a single processor, which limits scalability.
5. Robustness: Kernel-level threads are more robust as they can survive in the face of a thread failure. In contrast, a failure of a user-level thread can bring down the entire process.

10. (5%) What is the race condition?

Ans.

A race condition occurs when two or more threads or processes access a shared resource such as a variable, file, or database record, and the final outcome of the program depends on the order in which the threads or processes execute.

11. (6%) What are the internal and external fragmentations?

Ans.

Internal fragmentation occurs when a process or program is allocated more memory than it needs, and the unused portion of the memory remains allocated and unusable. This can occur when memory is allocated in fixed-size blocks, and the requested size of the memory allocation is smaller than the block size. The unused portion of the memory in the block is called internal fragmentation.

External fragmentation occurs when there is enough total memory available to satisfy a memory allocation request, but the memory is not contiguous or available in a single block. This can happen when memory is allocated and deallocated frequently, leaving small holes or gaps in the memory space that are too small to be used for a new allocation. These gaps can accumulate over time, resulting in a situation where there is enough free memory, but it is scattered in small chunks across the memory space, making it difficult to find a contiguous block of memory to satisfy a large memory allocation request.

12. (2%) The File Allocation Table of Microsoft's FAT32 file system is a variation of:

- (a) linked allocation.
- (b) contiguous allocation.
- (c) indexed allocation.
- (d) combined indexing.

Ans. A

13. (2%) Page fault frequency monitoring is a technique for:

- (a) evaluating the effectiveness of the MMU.
- (b) evaluating the effectiveness of the TLB.
- (c) managing the resident set for all processes in the system.
- (d) optimizing the interrupt response time for page faults.

Ans. C

14. (2%) Thrashing occurs when:

- (a) The scheduler flip-flops between two processes, leading to the starvation of others.
- (b) The sum of the working sets of all processes exceeds available memory.
- (c) Two or more processes compete for the same region of shared memory and wait on mutex locks.
- (d) Multiple processes execute in the same address space.

Ans. B

15. (2%) Large page sizes increase:

- (a) The working set size.
- (b) External fragmentation.
- (c) The page table size.
- (d) Internal fragmentation.

Ans. D

16. (2%) During a context switch, the operating system

- (a) Modifies some page table entries for the new process to reflect its memory map.
- (b) Switches a page table register to select a different page table.
- (c) Modifies the access permissions within the page table for the new process.
- (d) Leaves the page table untouched since it is a system-wide resource.

Ans. B

17. (2%) A semaphore puts a thread to sleep:

- (a) if it tries to decrement the semaphore's value below 0.
- (b) if it increments the semaphore's value above 0.
- (c) until another thread issues a notify on the semaphore.
- (d) until the semaphore's value reaches a specific number.

Ans. A

18. (2%) There are three processes on a system with the following performance requirements:

process A runs every 90 milliseconds, using 30 milliseconds of CPU time.

process B runs every 60 milliseconds, using 20 milliseconds of CPU time

process C runs every 500 milliseconds, using 10 milliseconds of CPU time

Use rate-monotonic analysis to assign priorities to the three processes.

Assume that priorities range from 0 to 90, with 0 being the highest. Which of the following is a valid set of priority assignments?

- (a) PA=40, PB=30, PC=100
- (b) PA=50, PB=40, PC=90
- (c) PA=70, PB=50, PC=60
- (d) PA=40, PB=60, PC=20

Ans. B

19. (2%) Which component of a process is not shared across threads?

- (a) Program memory.
- (b) Heap memory.
- (c) Global variables.
- (d) CPU Registers.

Ans. D

20. (2%) Compared to a non-preemptive scheduler, a preemptive scheduler can move processes from the:

- (a) running to the blocked state.
- (b) ready to the running state.
- (c) running to the ready state.
- (d) blocked to the ready state.

Ans. C

21. (2%) The reason for using a multilevel page table is to:

- (a) Reduce the amount of memory used for storing page tables.
- (b) Make table lookups more efficient than using a single-level table.
- (c) Make it easier to find unused page frames in the system.
- (d) Provide a hierarchy to manage different sections of a program.

Ans. A

22. (2%) A system has 32-bit logical addresses and 1 gigabyte main memory with page size of 256 Kbytes. How many entries of an inverted page table would contain:

- a. 1K entries
- b. 4K entries.
- c. 16K entries.
- d. 32K entries.

Ans. B

23. (2%) A system has 32-bit logical addresses and 1 gigabyte main memory with page size of 256 Kbytes. How many entries of a page table would contain:

- a. 1K entries
- b. 4K entries.

c. 16K entries.

d. 32K entries.

Ans. C

24. (2%) Which of these statements is true?

(a) A mode switch precedes a context switch.

(b) A context switch precedes a mode switch.

(c) A context switch can occur without a mode switch.

(d) A mode switch is just a different name for a context switch.

Ans. A